

MedCRP-CL: Continual Medical Image Segmentation via Bayesian Nonparametric Semantic Modality Discovery

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Abstract

Medical image segmentation faces a fundamental challenge in continual learning: data arrives sequentially from heterogeneous sources, yet effective continual learning requires discovering which tasks share sufficient structure to benefit from joint learning. Existing methods either apply uniform constraints across all tasks, causing catastrophic forgetting when tasks conflict, or require predefined task groupings that cannot anticipate future task diversity. We introduce MedCRP-CL, a framework that performs online task structure discovery and structure-aware continual learning. Leveraging the Chinese Restaurant Process (CRP), our method dynamically infers task groupings from clinical text prompts as tasks arrive, without requiring predefined cluster counts or access to future tasks. We term these discovered groupings semantic modalities, as they capture finer-grained structure than physical imaging modalities by integrating anatomical region and pathological context. Guided by this discovered structure, we maintain semantic modality-specific LoRA adapters regularized by intra-modality EWC, ensuring parameter isolation across dissimilar task groups while facilitating knowledge transfer within similar ones. The framework is also replay-free, storing only aggregate statistics rather than raw patient data. Experiments on 16 medical segmentation tasks across four imaging modalities demonstrate that MedCRP-CL achieves 73.3% Dice score with only 4.1% forgetting, outperforming the best baseline by 8.0% while requiring 6× fewer parameters. Code is available at <https://github.com/zygao930/MedCRP-CL>.

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1. Introduction

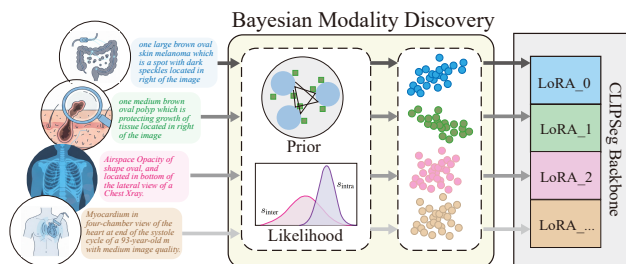


Figure 1. Overview of MedCRP-CL. Clinical prompts are assigned to semantic modalities via Bayesian inference combining CRP prior and adaptive likelihood.

Medical image segmentation is fundamental to clinical diagnosis and treatment planning, enabling quantitative analysis of anatomical structures and pathological regions (Poudel et al., 2024). In clinical practice, medical imaging data is routinely aggregated from heterogeneous sources including public benchmarks, multi-center trials, and partner institutions (Guan & Liu, 2022; Guan et al., 2024; Chang et al., 2023). This data arrives continuously from diverse clinical workflows: chest X-rays from emergency departments, ultrasound scans from cardiology, and endoscopic images from gastroenterology (Perkonig et al., 2021). Such sequential data acquisition creates a critical challenge for deployed segmentation models: they must continually adapt to new tasks while preserving performance on previously learned ones (Verma et al., 2023; Kumari et al., 2025). Vision-language segmentation models offer promising capabilities for this setting by leveraging natural language prompts to guide segmentation (Lüddecke & Ecker, 2022; Zhao et al., 2025), yet their deployment in continual learning scenarios remains largely unexplored.

Existing continual learning methods face a fundamental tension between parameter sharing and parameter isolation. Methods like EWC (Kirkpatrick et al., 2017) apply uniform parameter constraints across all tasks, forcing the model to compromise between conflicting objectives when tasks are dissimilar (Perkonig et al., 2021; McCloskey & Cohen, 1989). Mixture-of-experts approaches (Yu et al., 2024) attempt task-specific adaptation but require predefined ex-

pert counts that cannot anticipate future task diversity. The core issue is that indiscriminate sharing accelerates forgetting across dissimilar tasks, while rigid isolation precludes beneficial transfer among related tasks. Addressing this tension requires discovering the underlying task structure: which tasks are sufficiently similar to benefit from shared representations, and which require separate parameters.

Discovering such task structure in medical imaging is non-trivial. Physical imaging modality labels (e.g., “ultrasound”, “X-ray”), though readily available in clinical metadata, provide insufficient granularity: cardiac ultrasound and breast ultrasound share the same acquisition principles but involve fundamentally different anatomical structures and pathological patterns (Leclerc et al., 2019; Al-Dhabyani et al., 2020). Clustering based on raw image features is computationally expensive and unreliable due to high dimensionality and cross-site acquisition variability (Zhang et al., 2022). We observe that clinical text prompts offer a more effective basis for task grouping, as they naturally encode the combination of anatomical region and pathological context (Huang et al., 2021; Zhao et al., 2025). For instance, “chest X-ray showing pleural effusion” and “ultrasound of breast lesion” occupy distinct regions in prompt embedding space, reflecting their clinically meaningful differences.

Based on this observation, we present MedCRP-CL, a framework that discovers task groupings from clinical prompt embeddings to guide continual learning (Figure 1). We term these discovered groupings *semantic modalities* to distinguish them from physical imaging modalities, as they capture finer-grained structure combining anatomy and pathology. Given a sequence of tasks $\mathcal{T} = \{T_1, T_2, \dots, T_N\}$, our framework jointly learns: (1) a clustering function $z : \mathcal{T} \rightarrow \mathbb{N}$ assigning tasks to semantic modalities based on prompt embeddings, and (2) semantic modality-specific parameters θ_k enabling knowledge sharing within modalities while maintaining isolation across them. We employ the Chinese Restaurant Process (Blei et al., 2010) to automatically determine the number of semantic modalities, combined with semantic modality-specific LoRA adapters (Hu et al., 2022) regularized by intra-modality EWC.

Our contributions are threefold:

- We propose a **Bayesian nonparametric approach** for automatic semantic modality discovery in continual medical image segmentation. The CRP-based mechanism dynamically infers task groupings from clinical text prompts without requiring predefined labels or expert counts.
- We design a **semantic modality-aware continual learning architecture** that combines modality-specific LoRA adapters with intra-modality EWC regularization, achieving parameter isolation across semantic

modalities while enabling knowledge transfer within them. Our approach is also **replay-free**, eliminating the need to store historical patient data.

- We demonstrate strong performance on 16 medical segmentation tasks spanning four imaging modalities, outperforming the best baseline by 8.0% in Dice score while achieving only 4.1% forgetting and requiring $6\times$ fewer parameters. The discovered semantic modalities capture **finer-grained structure** than physical imaging modalities alone.

2. Related Work

Continual Learning for Vision-Language Models Continual learning aims to mitigate catastrophic forgetting during sequential task learning. Regularization-based methods, such as EWC (Kirkpatrick et al., 2017), penalize changes to critical parameters by estimating Fisher information. Replay-based approaches (Rebuffi et al., 2017; Chaudhry et al., 2019) store and rehearse past samples, but are impractical in medical settings where patient data cannot be retained due to privacy regulations such as HIPAA and GDPR. Recently, several works have extended continual learning to vision-language models. RAPF (Huang et al., 2024) combines representation adjustment with parameter fusion. CL-LoRA (He et al., 2025) applies low-rank adaptation alongside knowledge distillation. MoE-Adapters (Yu et al., 2024) employs mixture-of-experts routing for task-specific adaptation. However, these methods often assume homogeneous task distributions or require predefined expert counts, making them suboptimal for medical imaging where tasks involve heterogeneous modalities and unknown structures.

Continual Medical Image Segmentation Several recent methods address continual learning for medical image segmentation. MedPEFT-CL (Gao & Morel, 2026) introduces dual-phase parameter-efficient adaptation with bidirectional memory consolidation, but requires a replay buffer that conflicts with clinical privacy constraints. Low-Rank MoE (Chen et al., 2024) allocates an independent LoRA expert per task, avoiding forgetting at the cost of linear parameter growth and no knowledge sharing among semantically related tasks. FR²Seg (Xu et al., 2025) targets cross-site domain adaptation for a fixed segmentation task via Fourier-based transfer, rather than learning diverse tasks sequentially. Recent advances in structure-aware medical imaging such as HFF-Net (Shao et al., 2025b) and TRACE (Shao et al., 2025a) suggest promising directions for extending continual segmentation to 3D volumetric settings.

Task Structure Discovery in Continual Learning Existing continual learning methods typically assume tasks are either independent or share a global structure. Dynamic ar-

chitecture methods (Rusu et al., 2016; Yoon et al., 2018) expand network capacity for new tasks but lack mechanisms to identify shared structure. Task-agnostic approaches (Aljundi et al., 2019; Zeno et al., 2021) attempt to detect task boundaries automatically but still treat all tasks uniformly without grouping related ones. CAT (Ke et al., 2020) detects whether new tasks are similar or dissimilar to previous ones and applies different strategies accordingly, but relies on binary classification rather than discovering arbitrary cluster structures. Online clustering methods such as Online K-Means (MacQueen, 1967) and DP-Means (Kulis & Jordan, 2012) can dynamically assign data points to clusters, but require either a predefined number of clusters or sensitive distance thresholds. Bayesian nonparametric methods such as the Chinese Restaurant Process (Blei et al., 2010) offer principled frameworks for clustering with an unknown number of groups, yet their application to continual learning in medical imaging remains unexplored.

3. Method

Problem Formulation We consider continual learning for medical image segmentation, where a sequence of tasks $\mathcal{T} = \{T_1, T_2, \dots, T_N\}$ arrives sequentially from diverse clinical sources. Each task T_t consists of a dataset $\mathcal{D}_t = \{(x_i^t, y_i^t, p_i^t)\}_{i=1}^{n_t}$, where x_i^t denotes a medical image, y_i^t is the segmentation mask, and p_i^t is a clinical text prompt describing the target anatomical structure or pathology.

Medical images are acquired through distinct physical processes that induce fundamentally different visual distributions (Guan & Liu, 2022). We denote this latent structure as semantic modality partition $\mathcal{M} = \{M_1, \dots, M_K\}$, where K is unknown a priori. Tasks within the same semantic modality share visual characteristics and benefit from parameter sharing, while cross-modality tasks exhibit significant distribution shift that necessitates parameter isolation.

Our objectives are: (1) learn a semantic modality assignment function $z : \mathcal{T} \rightarrow \mathbb{N}$ that clusters tasks into semantic modalities without supervision, (2) minimize catastrophic forgetting through semantic modality-aware regularization, and (3) operate without storing raw patient data to comply with privacy regulations. The overall objective is:

$$\mathcal{L} = \sum_{t=1}^N \mathcal{L}_{\text{seg}}(T_t; \theta_{z(t)}) + \sum_{k=1}^K \Omega_k(\theta_k) \quad (1)$$

where $\theta_{z(t)}$ denotes semantic modality-specific parameters and Ω_k is the forgetting penalty applied only within semantic modality k . Figure 2 illustrates the overall architecture.

3.1. Bayesian Nonparametric Modality Discovery

Our framework discovers semantic modalities automatically without predefining their number. Physical modalities are

fixed by imaging hardware, but semantic modalities capture finer-grained clinical distinctions that emerge from the data. A Bayesian nonparametric formulation naturally accommodates this: new semantic modalities are instantiated when tasks exhibit novel characteristics, while related tasks consolidate into existing clusters. The discovery process combines a CRP prior over partitions with a likelihood based on prompt embeddings, as detailed below.

Chinese Restaurant Process Prior We employ the Chinese Restaurant Process (CRP) (Blei et al., 2010) as a prior over semantic modality assignments. For task T_t , the prior probability is:

$$P(z_t = k \mid z_{1:t-1}, \alpha) = \begin{cases} \frac{n_k}{t-1+\alpha} & k \in \{1, \dots, K_{t-1}\} \\ \frac{\alpha}{t-1+\alpha} & k = \text{new} \end{cases} \quad (2)$$

where n_k counts tasks assigned to semantic modality k and $\alpha > 0$ controls the propensity for new modalities.

The CRP prior is modulated by a likelihood term based on prompt similarity. Even when an existing semantic modality has accumulated many tasks, a new task will only join if its prompt embedding is sufficiently similar to the cluster centroid. This ensures that semantic coherence takes precedence over cluster size in modality assignment.

Prompt-Based Semantic Similarity We extract task representations from clinical prompts using a frozen text encoder ϕ from CLIP (Radford et al., 2021). For task T_t :

$$e_t = \frac{1}{|\mathcal{P}_t|} \sum_{p \in \mathcal{P}_t} \frac{\phi(p)}{\|\phi(p)\|_2} \quad (3)$$

where $\mathcal{P}_t$ denotes unique prompts in task t . Each semantic modality k maintains a running centroid μ_k , and the similarity is computed as $s_{t,k} = \langle e_t, \mu_k \rangle$. When task t is assigned to semantic modality k , the centroid is updated online:

$$\mu_k \leftarrow \frac{n_k - 1}{n_k} \mu_k + \frac{1}{n_k} e_t \quad (4)$$

This incremental update computes the exact running mean. Only the centroid vector is stored per semantic modality, avoiding storage of individual task embeddings. The centroid is not renormalized after updates. Its norm implicitly captures cluster coherence, with consistent embeddings maintaining norms close to unity and diverse assignments yielding smaller similarity scores.

Adaptive Similarity Distributions Rather than hand-tuning similarity thresholds, we learn them from data. We model similarity scores as conditionally Gaussian: $s \mid \text{same} \sim \mathcal{N}(\mu_{\text{intra}}, \sigma_{\text{intra}}^2)$ for tasks belonging to the same semantic modality, and $s \mid \text{diff} \sim \mathcal{N}(\mu_{\text{inter}}, \sigma_{\text{inter}}^2)$ for tasks

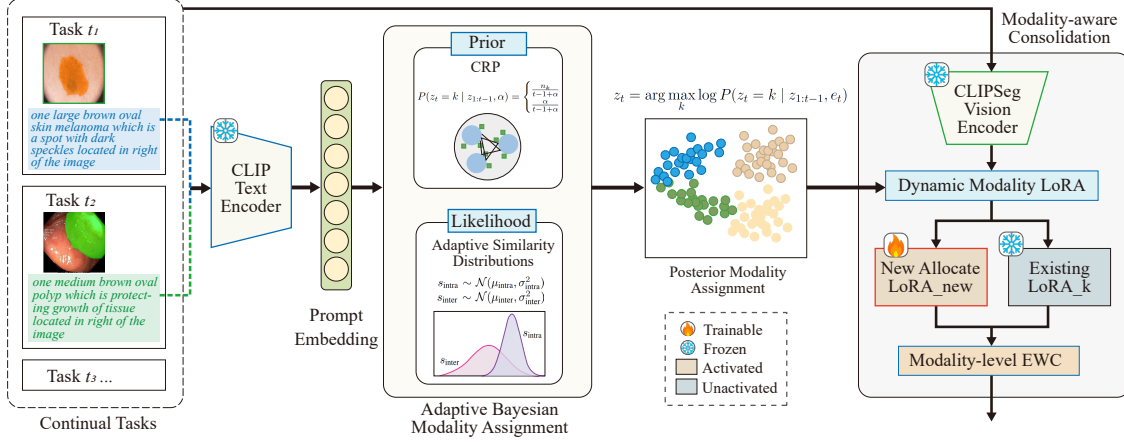


Figure 2. Overview of MedCRP-CL. For each incoming task, prompt embeddings are extracted and passed to the Bayesian modality assignment module, which combines CRP prior with learned similarity distributions to determine semantic modality membership. The assigned semantic modality’s LoRA adapter is then activated for training with intra-modality EWC regularization.

across different modalities. We maintain online estimates of both distributions, updated via Welford’s online algorithm (Welford, 1962). Given an observed similarity s between a new task and an existing semantic modality, this yields the log-likelihood ratio:

$$\ell(s) = \frac{(s - \mu_{\text{inter}})^2}{2\sigma_{\text{inter}}^2} - \frac{(s - \mu_{\text{intra}})^2}{2\sigma_{\text{intra}}^2} + \log \frac{\sigma_{\text{inter}}}{\sigma_{\text{intra}}} \quad (5)$$

To ensure numerical stability, we set a minimum standard deviation $\sigma_{\min} = 0.05$ for both distributions. During cold start when insufficient samples are available (fewer than one observation per distribution), we fall back to a logit-based likelihood $\ell(s) = \log(s + \epsilon) - \log(1 - s + \epsilon)$, treating similarity directly as a probability proxy. The cold-start logit mechanism maintains a similarity margin of 0.22+ between same-modality and cross-modality pairs, ensuring reliable assignment before Gaussian activation (see Appendix E.1). After Gaussian activation, the similarity distributions stabilize rapidly and maintain clear separation throughout training (see Appendix E.2). The distributions are updated as follows: when task t is assigned to modality k , the similarity $s_{t,k}$ updates the intra-modality distribution, while similarities $s_{t,j}$ for all $j \neq k$ update the inter-modality distribution. A positive $\ell(s)$ indicates that s is more consistent with intra-modality similarity, favoring assignment to the existing semantic modality.

Posterior Modality Assignment Combining prior and likelihood, the posterior for assigning task t to semantic modality k is:

$$\log P(z_t = k | z_{1:t-1}, e_t) = \log n_k - \log(t-1+\alpha) + \ell(s_{t,k}) \quad (6)$$

For a new semantic modality, letting $k^* = \arg \max_k s_{t,k}$:

$$\log P(z_t = \text{new}) = \log \alpha - \log(t-1+\alpha) - \ell(s_{t,k^*}) \quad (7)$$

The negative sign reflects that if the new task is dissimilar to all existing modalities (low $\ell(s_{t,k^*})$), it should form a new cluster. We perform MAP inference:

$$z_t = \arg \max_k \log P(z_t = k | z_{1:t-1}, e_t) \quad (8)$$

The likelihood term $\ell(s_{t,k})$ ensures that semantic similarity determines assignments. A task joins an existing semantic modality only when its prompt embedding is close to the cluster centroid; otherwise, a new semantic modality is created regardless of existing cluster sizes. Proposition A.1 guarantees near-zero assignment error under mild separation conditions (Appendix A).

3.2. Semantic Modality-Specific Continual Learning

Given semantic modality assignments, we instantiate semantic modality-specific parameters. Parameters are isolated across semantic modalities but shared among tasks within the same semantic modality.

Dynamic Semantic Modality-Specific LoRA We build upon a frozen vision-language backbone f_{Θ} (CLIPSeg) with semantic modality-specific low-rank adapters. For each linear layer $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, semantic modality k maintains:

$$W_k = W_0 + \frac{\alpha_{\text{LoRA}}}{r} B_k A_k \quad (9)$$

where $A_k \in \mathbb{R}^{r \times d_{\text{in}}}$, $B_k \in \mathbb{R}^{d_{\text{out}} \times r}$, and $r \ll d$. When a new semantic modality is discovered, fresh (A_k, B_k) pairs are allocated. This design achieves two goals: complete parameter isolation across semantic modalities prevents negative transfer between incompatible imaging types, while parameter sharing within each semantic modality enables positive transfer among related tasks.

Algorithm 1 MedCRP-CL: Adaptive CRP for Continual Medical Image Segmentation

Input: $\mathcal{T} = \{T_t\}_{t=1}^N$: Task sequence; $\phi(\cdot)$: Text encoder;
 α : CRP concentration; λ : EWC coefficient
Output: $\{\theta_k\}_{k=1}^K$: Semantic modality-specific LoRA parameters
 Initialize $\mathcal{M} \leftarrow \emptyset$; $(\mu_{\text{intra}}, \sigma_{\text{intra}}, \mu_{\text{inter}}, \sigma_{\text{inter}}) \leftarrow (0, 1, 0, 1)$
for each task $T_t = (\mathcal{D}_t, \mathcal{P}_t)$ in $\mathcal{T}$ **do**
 ▷ Semantic Modality Discovery
 Extract embedding $e_t \leftarrow \mathbb{E}_{p \sim \mathcal{P}_t} [\phi(p) / \|\phi(p)\|]$
 for each semantic modality k in $\mathcal{M}$ **do**
 Compute similarity $s_{t,k} \leftarrow \langle e_t, \mu_k \rangle$
 $\log P_k \leftarrow \log \frac{n_k}{t-1+\alpha} + \ell(s_{t,k})$
 end for
 $\log P_{\text{new}} \leftarrow \log \frac{\alpha}{t-1+\alpha} - \ell(\max_k s_{t,k})$
 $z_t \leftarrow \arg \max \log P$; allocate new LoRA if $z_t = \text{new}$
 ▷ Modality-Specific Training
 $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}} + \mathbf{1}_{[n_{z_t} > 1]} \cdot \Omega_{z_t}$
 Update θ_{z_t} until convergence
 ▷ Statistics Update
 Update μ_{z_t} and similarity distributions
 Compute $F_{z_t}^{(t)}$; update $\bar{F}_{z_t} \leftarrow \frac{n_{z_t}-1}{n_{z_t}} \bar{F}_{z_t} + \frac{1}{n_{z_t}} F_{z_t}^{(t)}$
 Store anchor parameters $\theta_{z_t}^* \leftarrow \theta_{z_t}$
end for
Return $\{\theta_k\}_{k=1}^{|\mathcal{M}|}$

Intra-Modality Elastic Weight Consolidation Parameter isolation across semantic modalities prevents cross-modality interference, but tasks within the same semantic modality still share parameters and may overwrite each other. We apply Elastic Weight Consolidation (EWC) to prevent this intra-modality forgetting. After training task t in semantic modality k , we estimate the Fisher information:

$$F_k^{(t)} = \mathbb{E}_{(x,y) \sim \mathcal{D}_t} [\nabla_{\theta_k} \log p(y|x; \theta_k)^{\otimes 2}] \quad (10)$$

The consolidated Fisher for semantic modality k is updated via exponential moving average:

$$\bar{F}_k \leftarrow \frac{n_k - 1}{n_k} \bar{F}_k + \frac{1}{n_k} F_k^{(t)} \quad (11)$$

For subsequent tasks in semantic modality k , the regularization term is:

$$\Omega_k(\theta_k) = \sum_i \bar{F}_{k,i} (\theta_{k,i} - \theta_{k,i}^*)^2 \quad (12)$$

EWC applies only within semantic modalities. Tasks in one semantic modality have no interaction with parameters of another, preventing conflicting gradients from incompatible visual distributions.

The training objective combines segmentation and regularization losses:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}} + \Omega_{z(t)}(\theta_{z(t)}) \quad (13)$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy loss, $\mathcal{L}_{\text{Dice}}$ addresses class imbalance in medical images, and $\Omega_{z(t)}$ regularizes only the current semantic modality's parameters. $\Omega_{z(t)} = 0$ for the first task in each semantic modality. Algorithm 1 summarizes the complete procedure.

4. Experiments

4.1. Experimental Setup

Datasets We evaluate on 16 medical image segmentation tasks spanning four imaging types and five anatomical regions: endoscopy (Jha et al., 2020; Bernal et al., 2015; Silva et al., 2014; Tajbakhsh et al., 2016; Vázquez et al., 2017), dermoscopy (Codella et al., 2018), ultrasound (Leclerc et al., 2019; Al-Dhabyani et al., 2020), and chest X-ray (Irvin et al., 2019). Text prompts are adopted from MedVLSM (Poudel et al., 2024). Table 1 summarizes the dataset statistics. Detailed dataset descriptions are provided in Appendix B.

Table 1. Dataset statistics for continual medical image segmentation.

Imaging Type	Organ	Dataset	# Train/Val/Test
Endoscopy	Colon	Kvasir	800/100/100
		ClinicDB	490/61/61
		ETIS	137/39/20
		CVC-300	64/21/23
		ColonDB	266/76/38
Dermoscopy	Skin	ISIC	810/90/379
Ultrasound	Heart	CAMUS	960/120/120
	Breast	BUSI-Benign	349/44/44
		BUSI-Malignant	168/21/21
X-ray	Chest	Airspace Opacity	75/17/33
		Atelectasis	48/9/23
		Cardiomegaly	35/14/19
		Edema	25/9/11
		Pleural Effusion	39/10/18
		Enlarged Cardio.	57/18/34
		Support Devices	59/17/31

Metrics We evaluate using two metrics: (1) **Average Dice Coefficient** measuring segmentation accuracy:

$$\text{Avg Dice} = \frac{1}{T} \sum_{i=1}^T \frac{2|P_i \cap G_i|}{|P_i| + |G_i|} \quad (14)$$

(2) **Average Forgetting Rate** quantifying knowledge retention:

$$\text{FR} = \frac{1}{T-1} \sum_{i=1}^{T-1} (\text{Dice}_i^{\text{peak}} - \text{Dice}_i^{\text{final}}) \quad (15)$$

where $\text{Dice}_i^{\text{peak}}$ is the best validation performance on task i observed immediately after training on that task, and $\text{Dice}_i^{\text{final}}$ is the performance after all tasks are learned. In Table 2, Params refers to trainable parameters, GPU denotes peak memory usage during training, and Time is relative to our method (1.0 $\times$).

Implementation Details We build upon the CLIPSeg architecture with a frozen backbone. LoRA modules (rank 8, $\alpha = 16$) are applied to the query, key, value, and output projections in both vision and text encoders; these adapters are trainable while the backbone parameters remain fixed. The text embeddings used for CRP modality assignment (Eq. 3) are extracted from the frozen CLIP text encoder before any LoRA adaptation, ensuring stable modality discovery throughout training. The CRP concentration parameter is set to $\alpha = 5.0$. For EWC, we use $\lambda = 5000$ with 200 samples for Fisher information estimation. Training uses AdamW with learning rate 1×10^{-3} and weight decay 8×10^{-5} . Each task trains for up to 60 epochs with early stopping (patience 8, minimum 15 epochs). All images are resized to 352×352 . Experiments are conducted on a single NVIDIA RTX 4090 GPU with batch size 16.

4.2. Comparison with State-of-the-Art Methods

Baselines We compare against four representative continual learning methods: EWC (Kirkpatrick et al., 2017), a regularization-based approach ($\lambda = 5000$); RAPF (Huang et al., 2024) and CL-LoRA (He et al., 2025), which employ parameter-efficient adapters with different fusion strategies; and MoE-Adapters (Yu et al., 2024), a mixture-of-experts approach ($K = 16$ experts with activate-freeze strategy). We also include *Sequential* fine-tuning without forgetting mitigation, and *Individual* models trained separately per task as an upper bound.

Overall Results Our method achieves the highest Dice score (73.3%) while maintaining the lowest forgetting rate (4.1%), significantly outperforming all baselines (Table 2). Compared to MoE-Adapters (Yu et al., 2024), which achieves 65.3% Dice with 51.9M parameters, our approach improves performance by 8.0% while using only 8.6M parameters—a 6 $\times$ reduction. Although CL-LoRA (He et al., 2025) employs minimal parameters (0.05M), it suffers from higher forgetting (9.7%) and requires 1.5 $\times$ more training time due to its knowledge distillation overhead. Classical regularization methods such as EWC (Kirkpatrick et al., 2017) and RAPF (Huang et al., 2024) show limited effectiveness in the medical imaging domain, achieving only 56.8% and 58.4% Dice respectively.

Figure 3 illustrates the performance dynamics across the 16-task sequence. Our method demonstrates remarkable stability: Cardiac US and Dermoscopy maintain near-constant

Table 2. Comparison of continual learning methods.

Method	Dice (%)	Forget (%)	Params (M)	GPU (GB)	Time
Individual	77.9	—	19.8	12.4	—
Sequential	48.0 $\pm$ 7.1	28.3 $\pm$ 7.7	1.2	5.8	0.7 $\times$
EWC	56.8 $\pm$ 3.7	11.3 $\pm$ 3.5	1.2	5.8	0.9 $\times$
RAPF	58.4 $\pm$ 1.7	7.2 $\pm$ 2.6	0.9	5.6	0.8 $\times$
CL-LoRA	60.7 $\pm$ 2.0	9.7 $\pm$ 1.4	0.05	5.7	1.5 $\times$
MoE-Adapters	65.3 $\pm$ 3.4	7.1 $\pm$ 3.2	51.9	13.3	1.2 $\times$
Ours	73.3 $\pm$ 1.0	4.1 $\pm$ 0.8	8.6	12.4	1.0 $\times$

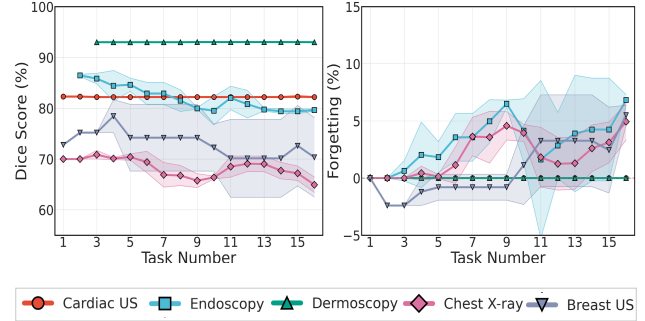


Figure 3. Performance analysis across all task orderings. Left: Dice score retention over the 16-task sequence. Right: Forgetting rate. Shaded regions indicate standard deviation across orderings. Negative forgetting indicates backward transfer.

Dice scores throughout training, while Endoscopy shows only modest degradation. Notably, the forgetting rates remain close to zero for most modalities, with occasional negative values indicating beneficial backward transfer. In contrast, Chest X-ray and Breast US exhibit higher variance, reflecting the inherent challenge of cross-modality continual learning. These results validate that our CRP-guided modality discovery effectively routes tasks to appropriate experts, preserving modality-specific knowledge while enabling positive knowledge transfer.

Order Sensitivity Analysis To evaluate the robustness to task ordering, we conduct experiments with four different task sequences: grouped (similar modalities consecutive), interleaved (alternating modalities), mixed (randomized), and reversed (inverse of grouped order). Full task sequences are provided in Appendix D. As shown in Figure 5, our method demonstrates consistent performance across all orderings, with Dice scores ranging from 0.72 to 0.74 and forgetting rates between 0.04 and 0.06. In contrast, MoE-Adapters exhibits greater sensitivity to task ordering, with Dice scores varying from 0.62 to 0.70 and higher forgetting rates (0.11-0.16). The narrow variability bands indicate effective adaptation to different task sequences, and the CRP-based semantic modality discovery mechanism does not require prior knowledge of optimal ordering.

Table 3. Comparison of different methods across datasets. Task order: CAMUS → Kvasir → ISIC → Airspace Op. → BUSI Ben. → ClinicDB → Atelectasis → ETIS → Cardiomeg. → CVC300 → BUSI Mal. → Edema → ColonDB → Enl. Cardio. → Pleural Eff. → Supp. Dev. Avg Dice shows segmentation performance; Avg FR (%) shows forgetting rate (lower is better). Bold indicates best performance among continual learning methods.

Method	CAMUS	Kvasir	ISIC	Airspace Op.	BUSI Ben.	ClinicDB	Atelectasis	ETIS	Cardiomeg.	CVC300	BUSI Mal.	Edema	ColonDB	Enl. Cardio.	Pleural Eff.	Supp. Dev.	Avg
<i>Dice (%) ↑</i>																	
Individual	83.4	86.6	93.1	72.1	83.6	84.4	67.8	70.8	73.3	92.3	72.6	77.4	76.5	72.3	66.7	73.1	77.9
Sequential	26.7	62.6	59.1	54.5	34.6	49.3	56.3	28.3	66.2	54.1	50.9	59.4	42.3	67.0	56.3	68.2	52.2
EWC	61.7	79.5	89.7	41.5	57.7	63.5	37.7	48.4	45.0	62.9	66.6	47.6	50.3	41.8	46.5	48.4	55.5
RAPF	17.3	75.8	87.0	68.8	57.9	58.4	65.4	40.1	68.9	49.1	69.3	70.9	44.4	66.7	54.0	72.0	60.4
CL-LoRA	37.4	66.4	84.6	68.6	53.3	56.5	66.8	30.7	70.9	61.1	69.4	74.6	50.3	68.8	66.7	72.1	62.4
MoE-Adapters	42.4	72.0	87.1	70.2	72.2	68.4	63.2	49.9	72.4	69.4	70.7	73.9	66.5	67.2	66.7	70.3	67.7
Ours	82.3	82.6	93.0	62.2	51.7	79.9	60.9	68.9	70.6	90.3	66.9	66.7	77.3	71.2	57.6	73.3	72.2
<i>Forgetting (%) ↓</i>																	
Sequential	59.4	24.8	33.7	13.8	45.9	36.0	10.5	37.6	5.1	41.8	24.2	14.9	32.2	3.2	0.0	0.0	23.9
EWC	24.6	5.9	3.0	17.8	16.4	10.6	25.5	3.2	10.3	18.8	6.5	6.7	11.1	6.8	32.4	26.6	14.1
RAPF	42.4	4.4	4.6	0.5	20.9	4.8	1.4	7.6	3.8	26.7	0.7	3.6	10.5	1.0	8.2	0.0	8.8
CL-LoRA	6.6	13.9	6.5	0.8	25.5	17.1	0.0	22.3	0.0	13.7	2.3	0.0	14.5	0.1	0.1	0.0	8.4
MoE-Adapters	40.7	12.6	5.5	0.0	3.8	12.7	0.3	10.9	0.6	11.6	0.8	0.1	4.7	1.3	0.0	0.0	6.6
Ours	0.0	3.7	0.0	7.6	14.0	3.1	1.4	4.8	4.7	2.2	0.0	13.0	0.0	1.4	11.4	0.0	4.2

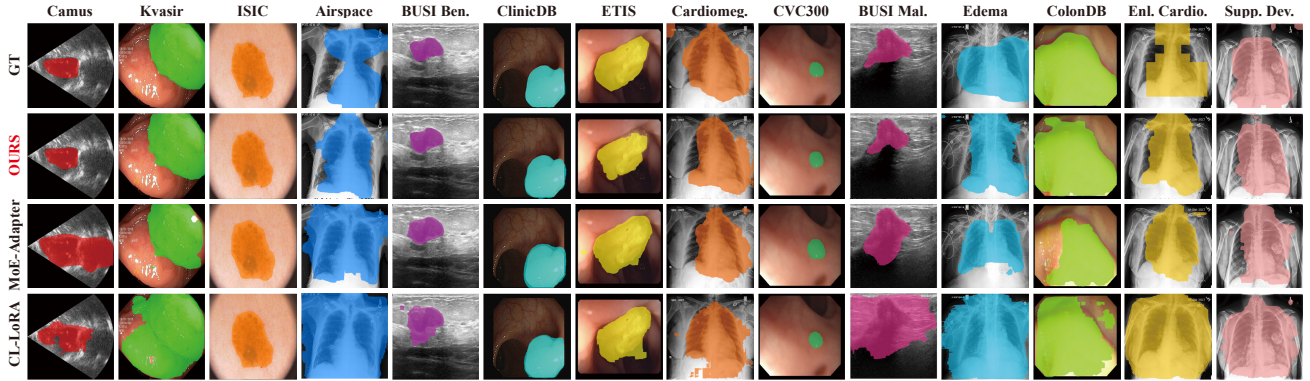


Figure 4. Visualization comparison of segmentation results cross representative tasks. Segmentation results from top row to bottom: Ground-truth, Ours, MoE-Adapters, and CL-LoRA.

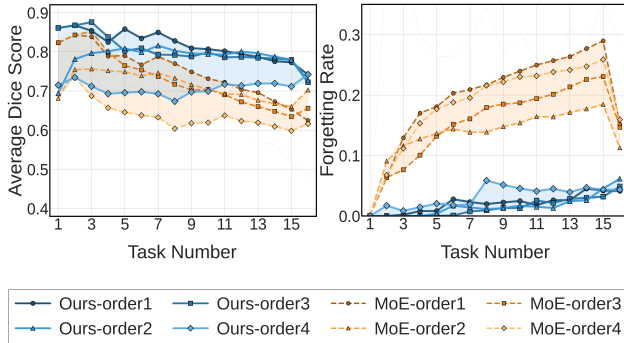


Figure 5. Order sensitivity analysis evaluating robustness to task ordering across four sequences. Our method (blue) maintains stable performance with narrow variability bands, while MoE-Adapters (orange) exhibits larger fluctuations.

Comparison on Interleaved Task Order Table 3 presents detailed per-task results on the mixed 16-task sequence. Our method achieves the best average Dice (72.2%) and lowest forgetting rate (4.2%), with strong performance on early tasks such as CAMUS (82.3%) and CVC300 (90.3%). This indicates effective knowledge retention throughout the learning process. While MoE-Adapters (Yu et al., 2024) and CL-LoRA (He et al., 2025) achieve competitive results on certain Chest X-ray tasks, they suffer from severe forgetting on others. Results on additional task orders are provided in Appendix F. Figure 4 provides qualitative comparisons: our method produces segmentation masks closest to ground truth, with sharper boundaries and fewer false positives, whereas baselines often exhibit incomplete regions or miss small structures.

4.3. Ablation Study

Module Ablation Analysis Table 4 evaluates each component’s contribution. Removing EWC increases forgetting from 4.09% to 5.41%, confirming that intra-modality regularization (Kirkpatrick et al., 2017) preserves knowledge within modality clusters. The most significant degradation occurs without CRP: forgetting rises to 15.55% as incompatible modalities interfere in a shared adapter. The Single LoRA baseline exhibits catastrophic forgetting (27.34%), consistent with naive fine-tuning. Removing LoRA yields near-zero forgetting but poor Dice (45.39%), as the frozen backbone cannot adapt to new tasks. These results confirm that CRP provides modality isolation, LoRA enables adaptation, and EWC consolidates intra-modality knowledge.

Table 4. Ablation study of different components.

Config.	CRP	LoRA	EWC	Dice	Forgetting
Full Model	✓	✓	✓	73.33	4.09
w/o EWC	✓	✓	×	71.92	5.41
w/o CRP	×	✓	✓	57.59	15.55
Single LoRA	×	✓	×	46.94	27.34
w/o LoRA	✓	×	✓	45.39	0.03

Loss Function Study Table 5 ablates our loss components. The full model ($\mathcal{L}_{Seg} + \mathcal{L}_{Dice} + \mathcal{L}_{EWC}$) achieves the best Dice (73.33%) with consistent modality discovery (5 clusters). Removing $\mathcal{L}_{EWC}$ increases forgetting from 4.09% to 5.25% and causes unstable modality allocation (5–6 clusters) as feature drift leads CRP to create redundant experts. Removing $\mathcal{L}_{Dice}$ achieves the lowest forgetting (3.66%) but sacrifices segmentation accuracy (71.87%), reflecting a stability-plasticity trade-off. Using only $\mathcal{L}_{Seg}$ yields the lowest performance (70.27%) with highly variable clustering (4–6 modalities). These results confirm that $\mathcal{L}_{EWC}$ stabilizes features for consistent CRP routing, while $\mathcal{L}_{Dice}$ directly optimizes segmentation quality.

Table 5. Ablation study of loss components.

Config.	Dice	Forgetting	#Mod.
$\mathcal{L}_{Seg}$	70.27	4.97	4–6
$\mathcal{L}_{Seg} + \mathcal{L}_{EWC}$	71.87	3.66	5
$\mathcal{L}_{Seg} + \mathcal{L}_{Dice}$	72.41	5.25	5–6
$\mathcal{L}_{Seg} + \mathcal{L}_{Dice} + \mathcal{L}_{EWC}$	73.33	4.09	5

4.4. Semantic Modality Discovery Analysis

Visualization To understand how our method organizes knowledge across tasks, we apply t-SNE to visualize the learned prompt embeddings (Figure 6). The visualization reveals that CRP automatically discovers five distinct semantic modality clusters, grouping tasks by their underlying imaging characteristics rather than their arrival order. No-

tably, CRP separates cardiac ultrasound (CAMUS) from breast ultrasound (BUSI), recognizing that these tasks require different feature representations despite sharing the same physical imaging modality. This emergent structure enables our method to route incoming tasks to appropriate experts, facilitating forward transfer within semantic modalities while isolating unrelated domains.

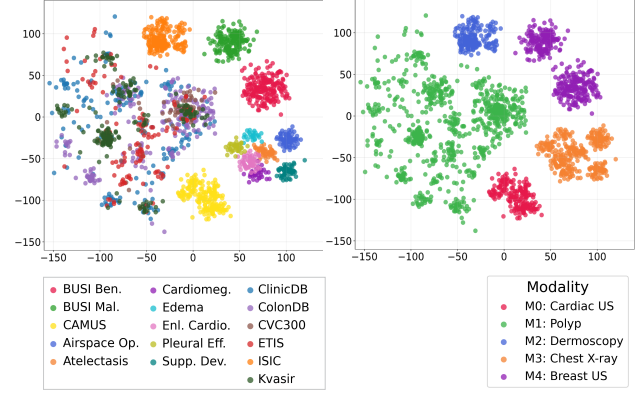


Figure 6. t-SNE visualization of prompt embeddings. Left: colored by dataset (16 classes). Right: colored by semantic modality discovered via CRP (5 clusters). Tasks with similar clinical semantics are automatically grouped together.

Comparison with Physical Modality Grouping Table 6 compares modality assignment strategies for continual learning. Grouping tasks by physical imaging modality, namely the acquisition device type (Ultrasound, Endoscopy, Dermoscopy, X-ray; $K=4$), merges cardiac and breast ultrasound into a single modality since both are acquired via ultrasound despite involving fundamentally different anatomical structures. This causes parameter interference and elevated forgetting (9.23%). Our CRP-based approach automatically discovers $K=5$ semantic modalities from clinical text prompts, correctly separating cardiac from breast ultrasound based on their distinct anatomical contexts, improving Dice by 7.6% and reducing forgetting by 5.1%. We further validate the necessity of this separation by merging all 10 cross-modality adapter pairs via Fisher-weighted averaging. All result in degradation (Appendix H).

Table 6. Impact of modality discovery on continual learning performance.

Modality Assignment	K	Dice	Forgetting
Physical imaging type	4	65.75	9.23
CRP discovered (Ours)	5	73.33	4.09

Text-Based vs. Visual-Based Clustering We compare text-only and visual-only clustering using embeddings from the same CLIP backbone. Table 7 both similarity statistics and discovered K across α values. Text embeddings pro-

vide a substantially larger intra-/inter-group gap (~ 0.50 vs. ~ 0.22 for visual), with the critical case being cardiac vs. breast ultrasound: visual similarity 0.95+ but text similarity ~ 0.45 . Consequently, text-only clustering discovers stable $K=5$ across all α , while visual-only and dual-track produce inconsistent K that shifts with α . Adding visual features degrades the correct $K=5$ structure that text-only achieves.

Table 7. Clustering signal comparison.

Embedding	Intra	Inter	Gap	Discovered K
Text-only	0.95+	0.45	~ 0.50	5
Visual-only	0.95+	0.73	~ 0.22	1–4

Robustness to Encoder Choice To verify that the discovered structure is not an artifact of the specific text encoder, we replace the CLIP encoder with 9 alternatives spanning 4 training paradigms, including medical contrastive models (Eslami et al., 2023; Zhang et al., 2023; luhuitong, 2024), general contrastive models (Cherti et al., 2023), and non-CLIP architectures combining contrastive learning with generative, masked modeling, or matching objectives (Singh et al., 2022; Li et al., 2022; Yu et al., 2022). As shown in Table 8, all 10 encoders discover identical $K=5$ with the same cluster membership at $\alpha=5$, confirming that the semantic modality structure is an intrinsic property of the data. Encoders differ only in embedding space compactness, which shifts the α range producing $K=5$: our default CLIP encoder maintains $K=5$ across the widest range.

We also tested two non-contrastive encoders, SigLIP and S-PubMedBERT, both of which produced $K=1$ across all α values due to insufficient inter-task separation in their embedding spaces. Our method requires text encoders trained with contrastive objectives.

Table 8. Encoder sensitivity analysis. All 10 encoders discover identical $K=5$ with the same cluster membership at $\alpha=5$.

Encoder	$\alpha=2$	$\alpha=5$	$\alpha=7$	$\alpha=10$
<i>Baseline</i>				
CLIP, CLIPSeg built-in (Ours)	5	5	5	5
<i>Medical VLP (contrastive, medical)</i>				
PubMedCLIP	5	5	5	5
BiomedCLIP	1	2	5	5
MedICaT-ROCO	5	5	5	10
<i>General VLP (contrastive, LAION)</i>				
OpenCLIP-B/32	2	5	5	5
OpenCLIP-B/16	2	5	5	5
OpenCLIP-L/14	4	5	5	5
<i>Non-CLIP (diverse training paradigms)</i>				
FLAVA	4	5	5	5
BLIP	3	5	5	5
CoCa	2	5	5	5

Prompt Robustness We evaluate prompt robustness at two levels. CRP discovers identical $K=5$ under both detailed prompts (e.g., “Left ventricular cavity of oval shape in four-chamber view...”) and concise prompts (e.g., “round polyp”), achieving 100% clustering consistency. Full prompt templates are in Appendix C. Table 9 evaluates robustness under clinically realistic perturbations including abbreviations, typos, keyword drops, and word reordering. All plausible noise conditions preserve $K=5$. Degradation occurs only under extreme perturbations ($>30\%$ typos or $>50\%$ keyword drop), where CRP defaults to increased parameter sharing rather than creating spurious clusters.

Table 9. CRP robustness to prompt perturbations. Clinically realistic perturbations (above the line) all preserve $K=5$.

Perturbation	Level	K
None (original)	–	5
Clinical abbreviation	–	5
Realistic typo	10–20%	5
Keyword drop	20–30%	5
Word shuffle	–	5
Realistic typo	30%	3
Keyword drop	50%	1
Generic prompt	–	1

5. Conclusion

We presented MedCRP-CL, a framework for continual medical image segmentation that automatically discovers semantic modality structure from clinical text prompts via the Chinese Restaurant Process. Combined with modality-specific LoRA adapters and intra-modality EWC regularization, our method achieves cross-modality isolation while enabling within-modality knowledge transfer. The framework is also replay-free, requiring no storage of raw patient data. Experiments demonstrate state-of-the-art performance with significantly reduced forgetting and fewer parameters. These results suggest that leveraging semantic information from clinical text prompts provides a principled approach to structure discovery in continual learning.

Impact Statement

This paper presents work whose goal is to advance continual learning for medical image segmentation. Our method is designed with privacy preservation as a core principle: by storing only aggregate statistics rather than raw patient data, it aligns with healthcare privacy regulations such as HIPAA and GDPR, potentially enabling deployment in privacy-sensitive clinical environments.

While improved medical image analysis could benefit clinical diagnosis and treatment planning, deployment in real clinical settings would require appropriate prospective vali-

dation and regulatory approval. We encourage practitioners to use such AI-assisted tools as decision support rather than replacements for clinical judgment. We do not foresee any immediate negative societal consequences specific to our methodological contributions beyond those common to the field of medical AI.

References

- Al-Dhabyani, W., Gomaa, M., Khaled, H., and Fahmy, A. Dataset of breast ultrasound images. *Data in Brief*, 28: 104863, 2020.
- Aljundi, R., Kelchtermans, K., and Tuytelaars, T. Task-free continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11254–11263, 2019.
- Bernal, J., Sánchez, F. J., Fernández-Esparrach, G., Gil, D., Rodríguez, C., and Vilariño, F. Wm-dova maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians. *Computerized Medical Imaging and Graphics*, 43:99–111, 2015.
- Blei, D. M., Griffiths, T. L., and Jordan, M. I. The nested Chinese Restaurant Process and Bayesian nonparametric inference of topic hierarchies. *Journal of the ACM*, 57(2): 1–30, 2010.
- Chang, Q., Yan, Z., Zhou, M., Qu, H., He, X., Zhang, H., Baskaran, L., Al’Aref, S., Li, H., Zhang, S., and Metaxas, D. N. Mining multi-center heterogeneous medical data with distributed synthetic learning. *Nature Communications*, 14(1):5510, 2023.
- Chaudhry, A., Rohrbach, M., Elhoseiny, M., Ajanthan, T., Dokania, P. K., Torr, P. H. S., and Ranzato, M. On tiny episodic memories in Continual Learning. *arXiv preprint arXiv:1902.10486*, 2019.
- Chen, Q., Zhu, L., He, H., Zhang, X., Zeng, S., Ren, Q., and Lu, Y. Low-rank mixture-of-experts for continual medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*, volume 15008 of *Lecture Notes in Computer Science*, pp. 382–392. Springer Nature Switzerland, 2024. doi: 10.1007/978-3-031-72111-3_36.
- Cherti, M., Beaumont, R., Wightman, R., Wortsman, M., Ilharco, G., Gordon, C., Schuhmann, C., Schmidt, L., and Jitsev, J. Reproducible scaling laws for contrastive language-image learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2818–2829, 2023.
- Codella, N. C. F., Gutman, D., Celebi, M. E., Helba, B., Marchetti, M. A., Dusza, S. W., Kallou, A., Liopyris, K., Mishra, N., Kittler, H., and Halpern, A. Skin lesion analysis toward melanoma detection: A challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), hosted by the International Skin Imaging Collaboration (ISIC). In *IEEE International Symposium on Biomedical Imaging*, pp. 168–172, 2018.
- Eslami, S., Meinel, C., and de Melo, G. Pubmedclip: How much does clip benefit visual question answering in the medical domain? In *Findings of the Association for Computational Linguistics: EACL 2023*, pp. 1181–1193, 2023.
- Gao, Z. and Morel, P. MedPEFT-CL: Dual-phase parameter-efficient continual learning with medical semantic adapter and bidirectional memory consolidation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 7699–7708, March 2026. doi: 10.1109/WACV61042.2026.00743.
- Guan, H. and Liu, M. Domain adaptation for medical image analysis: A survey. *IEEE Transactions on Biomedical Engineering*, 69(3):1173–1185, 2022.
- Guan, H., Yap, P.-T., Bozoki, A., and Liu, M. Federated learning for medical image analysis: A survey. *Pattern Recognition*, 151:110424, 2024.
- He, J., Duan, Z., and Zhu, F. CL-LoRA: Continual low-rank adaptation for rehearsal-free class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 30534–30544, 2025.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. LoRA: Low-rank adaptation of large language models. In *Proceedings of the International Conference on Learning Representations*, 2022.
- Huang, L., Cao, X., Lu, H., and Liu, X. Class-incremental learning with CLIP: Adaptive representation adjustment and parameter fusion. In *Proceedings of the European Conference on Computer Vision*, pp. 214–231. Springer, 2024.
- Huang, S.-C., Shen, L., Lungren, M. P., and Yeung, S. GLoRIA: A multimodal global-local representation learning framework for label-efficient medical image recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 3942–3951, 2021.
- Irvin, J., Rajpurkar, P., Ko, M., Yu, Y., Ciurea-Ilcus, S., Chute, C., Marklund, H., Haghighi, B., Ball, R., Shpanskaya, K., Seekins, J., Mong, D. A., Halabi, S. S., Sandberg, J. K., Jones, R., Larson, D. B., Langlotz, C. P., Patel, B. N., Lungren, M. P., and Ng, A. Y. Chexpert: A large chest radiograph dataset with uncertainty labels and

- expert comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pp. 590–597, 2019.
- Jha, D., Smedsrud, P. H., Riegler, M. A., Halvorsen, P., de Lange, T., Johansen, D., and Johansen, H. D. Kvasirseg: A segmented polyp dataset. In *MultiMedia Modeling (MMM)*, pp. 451–462. Springer, 2020.
- Ke, Z., Liu, B., and Huang, X. Continual learning of a mixed sequence of similar and dissimilar tasks. In *Advances in Neural Information Processing Systems*, volume 33, pp. 18493–18504, 2020.
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., and Hadsell, R. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.
- Kulis, B. and Jordan, M. I. Revisiting k-means: New algorithms via Bayesian nonparametrics. In *Proceedings of the 29th International Conference on International Conference on Machine Learning*, pp. 1131–1138, 2012.
- Kumari, P., Chauhan, J., Bozorgpour, A., Huang, B., Azad, R., and Merhof, D. Continual learning in medical image analysis: A comprehensive review of recent advancements and future prospects. *Medical Image Analysis*, 106: 103730, 2025.
- Leclerc, S., Smistad, E., Pedrosa, J., Østvik, A., Cervnansky, F., Espinosa, F., Espeland, T., Berg, E. A. R., Jodoin, P.-M., Grenier, T., Lartizien, C., D’hooge, J., Lovstakken, L., and Bernard, O. Deep learning for segmentation using an open large-scale dataset in 2d echocardiography. *IEEE Transactions on Medical Imaging*, 38(9):2198–2210, 2019.
- Li, J., Li, D., Xiong, C., and Hoi, S. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the 39th International Conference on Machine Learning*, pp. 12888–12900, 2022.
- Lüddecke, T. and Ecker, A. Image segmentation using text and image prompts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 7086–7096, 2022.
- luhuitong. CLIP-ViT-L-14-448px-MedICaT-ROCO. <https://huggingface.co/luhuitong/CLIP-ViT-L-14-448px-MedICaT-ROCO>, 2024.
- MacQueen, J. Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pp. 281–297, 1967.
- McCloskey, M. and Cohen, N. J. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24: 109–165, 1989.
- Perkonig, M., Hofmanninger, J., Herold, C. J., Brink, J. A., Panykh, O., Prosch, H., and Langs, G. Dynamic memory to alleviate catastrophic forgetting in continual learning with medical imaging. *Nature Communications*, 12(1): 5678, 2021.
- Poudel, K., Dhakal, M., Bhandari, P., Adhikari, R., Thapaliya, S., and Khanal, B. Exploring transfer learning in medical image segmentation using vision-language models. In *International Conference on Medical Imaging with Deep Learning*, pp. 1142–1165, 2024.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8748–8763. PMLR, 2021.
- Rebuffi, S.-A., Kolesnikov, A., Sperl, G., and Lampert, C. H. iCaRL: Incremental classifier and representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2001–2010, 2017.
- Rusu, A. A., Rabinowitz, N. C., Desjardins, G., Soyer, H., Kirkpatrick, J., Kavukcuoglu, K., Pascanu, R., and Hadsell, R. Progressive neural networks. *arXiv preprint arXiv:1606.04671*, 2016.
- Shao, M., Miao, X., Duan, H., Wang, Z., Chen, J., Huang, Y., Wu, X., Deng, J., Long, Y., and Zheng, Y. TRACE: Temporally reliable anatomically-conditioned 3d ct generation with enhanced efficiency. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, volume 15963 of *Lecture Notes in Computer Science*, pp. 627–637. Springer Nature Switzerland, 2025a. doi: 10.1007/978-3-032-04965-0_59.
- Shao, M., Wang, Z., Duan, H., Huang, Y., Zhai, B., Wang, S., Long, Y., and Zheng, Y. Rethinking brain tumor segmentation from the frequency domain perspective. *IEEE Transactions on Medical Imaging*, 44(11):4536–4553, 2025b. doi: 10.1109/TMI.2025.3579213.
- Silva, J., Histace, A., Romain, O., Dray, X., and Granado, B. Toward embedded detection of polyps in WCE images for early diagnosis of colorectal cancer. *International Journal of Computer Assisted Radiology and Surgery*, 9: 283–293, 2014.

- Singh, A., Hu, R., Goswami, V., Couairon, G., Galuba, W., Rohrbach, M., and Kiela, D. Flava: A foundational language and vision alignment model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15638–15650, 2022.
- Tajbakhsh, N., Gurudu, S. R., and Liang, J. Automated polyp detection in colonoscopy videos using shape and context information. *IEEE Transactions on Medical Imaging*, 35(2):630–644, 2016.
- Verma, T., Jin, L., Zhou, J., Huang, J., Tan, M., Choong, B. C. M., Tan, T. F., Gao, F., Xu, X., Ting, D. S., and Liu, Y. Privacy-preserving continual learning methods for medical image classification: A comparative analysis. *Frontiers in Medicine*, 10:1227515, 2023.
- Vázquez, D., Bernal, J., Sánchez, F. J., Fernández-Esparrach, G., López, A. M., Romero, A., Drozdal, M., and Courville, A. A benchmark for endoluminal scene segmentation of colonoscopy images. In *Journal of Healthcare Engineering*, volume 2017, pp. 4037190, 2017.
- Welford, B. P. Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3):419–420, 1962.
- Xu, C., Zhang, W., Zhang, H., Xu, X., Zhang, H., Zou, J., and Qin, J. FR²Seg: Continual segmentation across multiple sites via fourier style replay and adaptive consistency regularization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 8815–8823, 2025. doi: 10.1609/aaai.v39i8.32953.
- Yoon, J., Yang, E., Lee, J., and Hwang, S. J. Lifelong learning with dynamically expandable networks. In *Proceedings of the International Conference on Learning Representations*, 2018.
- Yu, J., Wang, Z., Vasudevan, V., Yeung, L., Seyedhosseini, M., and Wu, Y. Coca: Contrastive captioners are image-text foundation models. *Transactions on Machine Learning Research*, 2022.
- Yu, J., Zhuge, Y., Zhang, L., Hu, P., Wang, D., Lu, H., and He, Y. Boosting continual learning of vision-language models via mixture-of-experts adapters. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 23219–23230, 2024.
- Zeno, C., Golan, I., Hoffer, E., and Soudry, D. Task-agnostic continual learning using online variational Bayes with fixed-point updates. *Neural Computation*, 33(11):3139–3177, 2021.
- Zhang, M., Qu, L., Singh, P., Kalpathy-Cramer, J., and Rubin, D. L. SplitAVG: A heterogeneity-aware federated deep learning method for medical imaging. *IEEE Journal of Biomedical and Health Informatics*, 26(9):4635–4644, 2022.
- Zhang, S., Xu, Y., Usuyama, N., Xu, H., Bagga, J., Tinn, R., Preston, S., Rao, R., Wei, M., Valluri, N., Wong, C., Tupini, A., Wang, Y., Mazzola, M., Shukla, S., Liden, L., Gao, J., Crabtree, A., Piening, B., Bifulco, C., Lungren, M. P., Naumann, T., Wang, S., and Poon, H. Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:2303.00915*, 2023.
- Zhao, Z., Zhang, Y., Wu, C., Zhang, X., Zhou, X., Zhang, Y., Wang, Y., and Xie, W. Large-vocabulary segmentation for medical images with text prompts. *npj Digital Medicine*, 8(1):566, 2025.

A. Theoretical Analysis

We provide theoretical guarantees for our approach.

Proposition A.1 (Modality Clustering Consistency). *Assume intra-modality similarities follow $\mathcal{N}(\mu_{\text{intra}}, \sigma_{\text{intra}}^2)$ and inter-modality similarities follow $\mathcal{N}(\mu_{\text{inter}}, \sigma_{\text{inter}}^2)$ with separation $\Delta = \mu_{\text{intra}} - \mu_{\text{inter}} > 0$. If $\Delta > 2(\sigma_{\text{intra}} + \sigma_{\text{inter}})$, then MAP modality assignment achieves zero clustering error as $t \rightarrow \infty$ with probability 1.*

Proof. The log-likelihood ratio $\ell(s)$ defined in Eq. 5 acts as a binary classifier between same-modality and different-modality hypotheses. Under the Gaussian assumption, the optimal decision boundary is at:

$$s^* = \frac{\mu_{\text{intra}}\sigma_{\text{inter}}^2 + \mu_{\text{inter}}\sigma_{\text{intra}}^2}{\sigma_{\text{intra}}^2 + \sigma_{\text{inter}}^2} \quad (16)$$

The classification error is bounded by:

$$P_{\text{error}} = P(s_{\text{intra}} < s^*) + P(s_{\text{inter}} > s^*) \leq 2 \exp\left(-\frac{\Delta^2}{8(\sigma_{\text{intra}}^2 + \sigma_{\text{inter}}^2)}\right) \quad (17)$$

by the Chernoff bound. The separation condition $\Delta > 2(\sigma_{\text{intra}} + \sigma_{\text{inter}})$ ensures $P_{\text{error}} \rightarrow 0$ exponentially.

By the strong law of large numbers, Welford’s online estimates converge almost surely: $\hat{\mu}_{\text{intra}} \xrightarrow{a.s.} \mu_{\text{intra}}$ and $\hat{\sigma}_{\text{intra}} \xrightarrow{a.s.} \sigma_{\text{intra}}$ (similarly for inter-modality statistics). Combined with the CRP’s exchangeability property and de Finetti’s theorem, the posterior concentrates on the true partition as $t \rightarrow \infty$. $\square$

Proposition A.2 (Modality-Isolated Forgetting Bound). *Let $\mathcal{L}_t^*$ be the optimal loss achievable on task t . Under modality-isolated EWC with coefficient λ , the forgetting after training m subsequent tasks within the same modality satisfies:*

$$\mathcal{L}_t(\theta_k^{(m)}) - \mathcal{L}_t^* \leq \frac{1}{\lambda \cdot \lambda_{\min}(\bar{F}_k)} \sum_{j=1}^m \|\nabla_{\theta_k} \mathcal{L}_{t+j}\|^2 \quad (18)$$

where $\lambda_{\min}(\bar{F}_k)$ is the minimum eigenvalue of the consolidated Fisher matrix.

Proof. Following the analysis of Kirkpatrick et al. (2017), the EWC penalty constrains parameter movement in directions important for previous tasks. For task t in modality k , let $\theta_k^{(0)}$ be the parameters after training task t (which achieves $\mathcal{L}_t^*$), and $\theta_k^{(m)}$ be parameters after training m subsequent tasks.

The EWC objective ensures:

$$\|\theta_k^{(m)} - \theta_k^{(0)}\|_{\bar{F}_k}^2 \leq \frac{1}{\lambda} \sum_{j=1}^m \mathcal{L}_{t+j}(\theta_k^{(j-1)}) \quad (19)$$

Since $\theta_k^{(0)}$ minimizes $\mathcal{L}_t$, we have $\mathcal{L}_t(\theta_k^{(0)}) = \mathcal{L}_t^*$ and $\nabla \mathcal{L}_t(\theta_k^{(0)}) = 0$. By Taylor expansion:

$$\mathcal{L}_t(\theta_k^{(m)}) - \mathcal{L}_t^* \approx \frac{1}{2} (\theta_k^{(m)} - \theta_k^{(0)})^\top H_t (\theta_k^{(m)} - \theta_k^{(0)}) \quad (20)$$

Using $H_t \preceq \bar{F}_k$ (Fisher information approximates Hessian at convergence) and $\|x\|_{\bar{F}_k}^2 \geq \lambda_{\min}(\bar{F}_k) \|x\|^2$:

$$\mathcal{L}_t(\theta_k^{(m)}) - \mathcal{L}_t^* \leq \frac{1}{2\lambda_{\min}(\bar{F}_k)} \|\theta_k^{(m)} - \theta_k^{(0)}\|_{\bar{F}_k}^2 \leq \frac{1}{\lambda \cdot \lambda_{\min}(\bar{F}_k)} \sum_{j=1}^m \|\nabla_{\theta_k} \mathcal{L}_{t+j}\|^2 \quad (21)$$

Critically, this bound involves only tasks within modality k . Tasks in other modalities use separate parameters and do not contribute to forgetting on task t . $\square$

Implications. Proposition A.1 guarantees that our modality discovery mechanism correctly identifies the underlying structure given sufficient separation in prompt embeddings. In our experiments, we observe $\mu_{\text{intra}} \approx 0.94$ and $\mu_{\text{inter}} \approx 0.51$ with $\sigma_{\text{intra}} \approx 0.05$ and $\sigma_{\text{inter}} \approx 0.10$, yielding $\Delta \approx 0.43$ which substantially exceeds the required separation of $2(\sigma_{\text{intra}} + \sigma_{\text{inter}}) = 0.30$. Proposition A.2 shows that forgetting scales only with tasks within a modality, not the total task count—a key advantage over global EWC approaches.

B. Datasets Description

Kvasir-SEG (2020) (Jha et al., 2020): Endoscopic polyp segmentation dataset containing 1000 colonoscopy images with pixel-level annotations. Images feature diverse polyp morphologies, sizes, and locations within the colon, representing real clinical scenarios for computer-aided polyp detection during routine colonoscopy procedures.

ClinicDB (2015) (Bernal et al., 2015): Clinical endoscopic database for polyp segmentation comprising 612 colonoscopy frames with corresponding ground truth masks. Dataset emphasizes challenging cases with varying illumination conditions, polyp textures, and anatomical backgrounds commonly encountered in clinical practice.

ETIS (2014) (Silva et al., 2014): Endoscopic polyp segmentation dataset with 196 high-resolution colonoscopy images. Dataset emphasizes challenging polyp detection scenarios including flat lesions, small polyps, and cases with poor visibility conditions that test segmentation algorithm robustness in clinical environments.

ColonDB (2016) (Tajbakhsh et al., 2016): Comprehensive colonoscopy database containing 380 images with polyp segmentation masks. Dataset includes diverse polyp types, anatomical locations, and imaging conditions, providing extensive coverage of endoscopic appearance variations encountered during routine colonoscopic examinations.

CVC300 (2017) (Vázquez et al., 2017): Colonoscopy video database comprising 60 polyp segmentation cases extracted from endoscopic sequences. Dataset focuses on temporal consistency and motion artifacts in video-based polyp detection, essential for real-time clinical applications during live colonoscopy procedures.

ISIC 2016 (2016) (Codella et al., 2018): International Skin Imaging Collaboration dataset containing 1279 dermoscopy images for melanoma segmentation. Dataset includes diverse skin lesion types, pigmentation patterns, and imaging artifacts, representing global dermatological imaging standards for automated skin cancer detection.

CAMUS (2019) (Leclerc et al., 2019): Cardiac ultrasound segmentation dataset containing 6000 echocardiographic images with left ventricle annotations. Dataset covers multiple cardiac views and pathological conditions, enabling comprehensive evaluation of automated cardiac function assessment in clinical echocardiography.

BUSI (2020) (Al-Dhabyani et al., 2020): Breast ultrasound segmentation dataset comprising 780 images with lesion annotations. Dataset includes benign and malignant breast masses with varying echogenicity patterns, supporting development of automated breast cancer screening and diagnostic assistance tools. We split this dataset by pathology into BUSI-Benign and BUSI-Malignant subsets to enable fine-grained evaluation across lesion types.

CheXlocalize (2019) (Irvin et al., 2019): Chest X-ray pathology localization dataset containing 2177 radiographs with bounding box annotations. Dataset covers multiple thoracic pathologies including pneumonia, effusions, and nodules, enabling evaluation of automated radiological interpretation and diagnosis assistance systems. We create seven pathology-specific subsets (Airspace Opacity, Atelectasis, Cardiomegaly, Edema, Enlarged Cardiomediastinum, Pleural Effusion, and Support Devices) to evaluate segmentation performance on individual clinical findings.

C. Prompt Templates

Table 10. Prompt templates for all 16 datasets, organized by discovered semantic modality.

Dataset	Concise	Detailed
<i>Semantic Modality 0: Cardiac Ultrasound</i>		
CAMUS	Left ventricular cavity in the cardiac ultrasound.	Left ventricular cavity of oval shape in four-chamber view in the cardiac ultrasound at end of the diastole cycle of a 18-year-old male with good image quality.
<i>Semantic Modality 1: Endoscopy (Polyp Segmentation)</i>		
Kvasir-SEG	round polyp	One medium pink round polyp which is a projecting growth of tissue located in top left of the image.
ClinicDB	oval polyp	Two large white oval polyp which is a projecting growth of tissue located in center, bottom right of the image.
CVC-300	triangular polyp	One small brown triangular polyp which is a projecting growth of tissue located in bottom right of the image.
ColonDB	kidney polyp	One large orange kidney polyp which is a projecting growth of tissue located in center of the image.
ETIS	circle polyp	Two small yellow circle polyp which is a projecting growth of tissue located in top, right of the image.
<i>Semantic Modality 2: Dermoscopy (Skin Lesion Segmentation)</i>		
ISIC	circle skin melanoma	One large blue circle skin melanoma which is a dark sore with irregular texture located in center of the image.
<i>Semantic Modality 3: Chest X-ray</i>		
CheX-Airspace Opacity	Airspace Opacity in a chest Xray.	Airspace Opacity in a Chest Xray. Enlarged Cardiome-diastinum, Cardiomegaly, Lung Opacity are present.
CheX-Atelectasis	Atelectasis in a chest Xray.	Atelectasis in a Chest Xray. Enlarged Cardiome-diastinum, Lung Opacity, Atelectasis, Pleural Effusion are present.
CheX-Cardiomegaly	Cardiomegaly in a chest Xray.	Cardiomegaly in a Chest Xray. Enlarged Cardiome-diastinum, Cardiomegaly, Lung Opacity, Edema are present.
CheX-Edema	Edema in a chest Xray.	Edema in a Chest Xray. Lung Opacity, Edema, Consoli-dation, Atelectasis are present.
CheX-Enlarged Card.	Enlarged Cardiome-diastinum in a chest Xray.	Enlarged Cardiome-diastinum in a Chest Xray. Enlarged Cardiome-diastinum, Cardiomegaly, Lung Opacity are present.
CheX-Pleural Effusion	Pleural Effusion in a chest Xray.	Pleural Effusion in a Chest Xray. Lung Opacity, Atelecta-sis, Pleural Effusion, Support Devices are present.
CheX-Support Devices	Support Devices in a chest Xray.	Support Devices in a Chest Xray. Enlarged Cardiome-diastinum, Cardiomegaly, Support Devices are present.
<i>Semantic Modality 4: Breast Ultrasound</i>		
BUSI-Benign	Benign tumor in the breast ultrasound image.	Five small, medium circle-shaped benign tumors at the top right, top, center, right, right in the breast ultrasound image.
BUSI-Malignant	Malignant tumor in the breast ultrasound image.	One large irregular-shaped malignant tumor at the center in the breast ultrasound image.

D. Task Orders

To evaluate robustness to task arrival sequences, we test four distinct orderings spanning 16 segmentation tasks. Table 11 shows all orderings used in our experiments.

Table 11. Task orderings used in experiments. Each order presents 16 medical image segmentation tasks in a different sequence.

Order	Task Sequence
grouped	CAMUS → Kvasir → ClinicDB → ETIS → CVC-300 → ColonDB → ISIC → CheX-Airspace → CheX-Atelectasis → CheX-Cardiomegaly → CheX-Edema → CheX-Enlarged → CheX-Pleural → CheX-Support → BUSI-Benign → BUSI-Malignant
<i>Rationale</i>	<i>Tasks grouped by semantic modality, simulating sequential adoption of imaging equipment in a clinical setting.</i>
reversed	BUSI-Malignant → BUSI-Benign → CheX-Support → CheX-Pleural → CheX-Enlarged → CheX-Edema → CheX-Cardiomegaly → CheX-Atelectasis → CheX-Airspace → ISIC → ColonDB → CVC-300 → ETIS → ClinicDB → Kvasir → CAMUS
<i>Rationale</i>	<i>Reverse of grouped order to evaluate sensitivity to arrival direction while preserving modality clustering.</i>
interleaved	CAMUS → Kvasir → ISIC → CheX-Airspace → BUSI-Benign → ClinicDB → CheX-Atelectasis → ETIS → CheX-Cardiomegaly → CVC-300 → BUSI-Malignant → CheX-Edema → ColonDB → CheX-Enlarged → CheX-Pleural → CheX-Support
<i>Rationale</i>	<i>Alternating modalities to simulate realistic clinical deployment where new imaging types arrive unpredictably.</i>
mixed	CheX-Airspace → Kvasir → CAMUS → BUSI-Benign → ClinicDB → CheX-Cardiomegaly → ISIC → CheX-Atelectasis → ETIS → BUSI-Malignant → CheX-Edema → CVC-300 → ColonDB → CheX-Pleural → CheX-Enlarged → CheX-Support
<i>Rationale</i>	<i>Randomized task arrival to stress-test modality discovery under arbitrary task sequences.</i>

E. CRP Assignment Reliability

E.1. Cold-Start Assignment Trace

Before Gaussian activation, the CRP relies on a parameter-free logit-based likelihood with 0.5 as the decision boundary. Table 12 shows the first four task assignments under four different orderings. Same-modality pairs consistently produce similarity >0.79 while cross-modality pairs produce similarity <0.57 , providing a margin of 0.22+ for reliable separation. All pre-Gaussian assignments are consistent with the final converged $K=5$ structure across all orderings.

Table 12. Per-task assignment trace during cold-start (first 4 tasks) under 4 orderings. Values indicate similarity logits; JOIN/NEW denotes the CRP decision.

Order	Task 1	Task 2	Task 3	Task 4
Grouped	Kvasir NEW (0.474)	ClinicDB JOIN-M1 (0.991)	ETIS JOIN-M1 (0.980)	CVC300 JOIN-M1 (0.992)
Interleaved	Kvasir NEW (0.474)	ISIC NEW (0.548)	ChexOp. NEW (0.488)	BUSI-B. NEW (0.568)
Mixed	Kvasir NEW (0.305)	CAMUS NEW (0.488)	BUSI-B. NEW (0.568)	ClinicDB JOIN-M1 (0.991)
Reversed	BUSI-B. JOIN-M0 (0.915)	ChexSup. NEW (0.493)	ChexEnl. JOIN-M1 (0.790)	ChexEdema JOIN-M1 (0.870)

E.2. Gaussian Estimate Stabilization

Table 13 tracks the intra- and inter-modality Gaussian statistics as observations accumulate. The intra/inter gap remains above 0.45 throughout and stabilizes within 2–3 observations after activation. Final distributions are consistent across all 4 task orderings (intra_mean: 0.921–0.931, inter_mean: 0.429–0.473).

Table 13. Gaussian estimate convergence. The intra/inter similarity gap stabilizes within 2–3 observations.

Intra n	μ_{intra}	σ_{intra}	μ_{inter}	Gap
1	0.980	0.100	0.445	0.535
2	0.925	0.078	0.448	0.477
3	0.947	0.067	0.444	0.503
8	0.921	0.064	0.429	0.492

F. Results Across Different Task Orderings

Table 14. MedCRP-CL performance on grouped order: CAMUS → Kvasir → ClinicDB → ETIS → CVC300 → ColonDB → ISIC → Airspace Op. → Atelectasis → Cardiomeg. → Edema → Enl. Cardio. → Pleural Eff. → Supp. Dev. → BUSI Ben. → BUSI Mal.

	CAMUS	Kvasir	ClinicDB	ETIS	CVC300	ColonDB	ISIC	Airspace Op.	Atelectasis	Cardiomeg.	Edema	Enl. Cardio.	Pleural Eff.	Supp. Dev.	BUSI Ben.	BUSI Mal.	Avg
Dice (%) ↑	82.3	82.6	79.9	68.9	90.3	77.3	93.0	62.3	60.8	69.6	66.7	71.0	57.7	73.8	76.8	74.7	74.2
Fgt. (%) ↓	0.0	3.7	3.1	4.8	2.2	0.0	0.0	7.4	0.5	5.4	12.9	1.5	9.4	0.0	3.3	0.0	3.4

Table 15. MedCRP-CL performance on mixed order: Airspace Op. → Kvasir → CAMUS → BUSI Ben. → ClinicDB → Cardiomeg. → ISIC → Atelectasis → ETIS → BUSI Mal. → Edema → CVC300 → ColonDB → Pleural Eff. → Enl. Cardio. → Supp. Dev.

	Airspace Op.	Kvasir	CAMUS	BUSI Ben.	ClinicDB	Cardiomeg.	ISIC	Atelectasis	ETIS	BUSI Mal.	Edema	CVC300	ColonDB	Pleural Eff.	Enl. Cardio.	Supp. Dev.	Avg
Dice (%) ↑	57.8	82.6	82.0	76.6	78.8	68.3	93.0	56.4	67.9	75.3	62.9	90.0	77.3	51.9	69.4	73.2	72.7
Fgt. (%) ↓	12.2	4.2	0.0	5.1	5.0	5.9	0.0	5.1	4.9	0.0	17.0	2.1	0.0	16.4	4.0	0.0	5.1

Table 16. MedCRP-CL performance on reversed order: BUSI Mal. → BUSI Ben. → Supp. Dev. → Enl. Cardio. → Edema → Cardiomeg. → Atelectasis → Airspace Op. → ISIC → ColonDB → CVC300 → ETIS → ClinicDB → Kvasir → Pleural Eff. → CAMUS.

	BUSI Mal.	BUSI Ben.	Supp. Dev.	Enl. Cardio.	Edema	Cardiomeg.	Atelectasis	Airspace Op.	ISIC	ColonDB	CVC300	ETIS	ClinicDB	Kvasir	Pleural Eff.	CAMUS	Avg
Dice (%) ↑	71.1	79.2	65.0	63.7	73.0	65.9	56.2	69.4	93.1	72.9	87.6	66.2	80.2	86.8	65.0	82.6	73.6
Fgt. (%) ↓	1.7	0.0	6.7	7.6	4.5	9.4	10.9	2.4	0.0	4.0	6.1	3.5	3.4	0.0	3.0	0.0	3.9

G. Prompt Perturbation Examples

Table 17 provides the full noisy prompt examples used in the prompt robustness evaluation (Table 9). All perturbations are applied to the same original prompt describing a cardiac ultrasound task: “Left ventricular cavity of rectangle shape in two-chamber view of the heart at end of the diastole cycle of a 56-year-old f with good image quality.”

Table 17. Prompt perturbation examples corresponding to Table 9.

Perturbation	Noisy Example
<i>Clinically Realistic</i>	
None (original)	Left ventricular cavity of rectangle shape in two-chamber view of the heart at end of the diastole cycle of a 56-year-old f with good image quality.
Abbreviation	Left ventricular cavity of rectangle shape in 2CV of the heart at ED of a 56-year-old f with good IQ.
Typo 10%	Left ventricular cavity of rectangle shape in two-chamber veiw of the heart at end of the diastole cycle of a 56-year-old f with good image quality.
Typo 20%	Left ventricular cavit of rectangle sshape in two-chambr vie of the heart at end of the diastole cycle of a 56-year-old f with good image quality.
Drop 20%	Left ventricular cavity of rectangle shape in two-chamber view the heart end of the cycle of 56-year-old f with good image.
Drop 30%	Left ventricular cavity of rectangle two-chamber view the heart end of the cycle of 56-year-old f with good image.
Shuffle	diastole at of quality. a in shape the the with heart of ventricular of good end cavity cycle image rectangle f two-chamber view Left of 56-year-old.
<i>Extreme Perturbations</i>	
Typo 30%	Left venricular cavit of rectangle sshape in two-chambr vie of tthe heart at ennd of tge diastle cycle of a 56-year-old f with good image quality.
Drop 50%	Left cavity of rectangle two-chamber view end of cycle 56-year-old f good image.
Generic	segment the region.

H. Fisher-Weighted Merge Analysis

We merge each pair of modality-specific LoRA adapters using Fisher-weighted averaging, with Fisher information matrices already computed during EWC. After merging, all affected tasks are re-adapted for 5 epochs until convergence. Results are shown in Table 18. All 10 pairs degrade, confirming that each discovered semantic modality captures distinct representations that cannot be consolidated without loss.

Table 18. Fisher-weighted merge results across all 10 cross-modality pairs. All merges degrade performance despite re-adaptation.

Merge Pair	Modalities	Before	After	Δ
M0+M1	Cardiac+Polyp	0.793	0.398	−0.395
M0+M2	Cardiac+Dermoscopy	0.871	0.643	−0.228
M0+M3	Cardiac+ChestXR	0.686	0.538	−0.148
M0+M4	Cardiac+BrestUS	0.772	0.457	−0.315
M1+M2	Polyp+Dermoscopy	0.812	0.654	−0.158
M1+M3	Polyp+ChestXR	0.718	0.544	−0.174
M1+M4	Polyp+BrestUS	0.778	0.546	−0.232
M2+M3	Dermoscopy+ChestXR	0.701	0.685	−0.016
M2+M4	Dermoscopy+BrestUS	0.812	0.754	−0.058
M3+M4	ChestXR+BrestUS	0.687	0.647	−0.039